

FH Aachen

Department of Electrical Engineering and Information Technology

Exposé

**Evaluating AI Agent Capabilities in SysML v2 Modeling
via an MCP Bridge to Magic Systems of Systems Architect**

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1 Exposé

Motivation

[TODO 150 words]

Problem Statement

[TODO 150 words]

Research Question and Hypotheses

RQ: Which SysML v2 modeling tasks can an LLM-based agent perform reliably in an industrial modeling environment (MSoSA) when connected through an MCP bridge, and where do its capabilities break down?

- **H1 (Task asymmetry):** Read-only tasks (Abfragen, Validierung) are performed with high reliability, whereas write tasks (Korrektur, Erstellen) and Verifizierung degrade with model size and cross-layer dependency depth.
- **H2 (Tool semantics):** Exposing tool-native capabilities (validation suites, expression evaluation) as MCP tools yields larger gains than a thin CRUD wrapper over the SysML v2 API alone.
- **H3 (Model scale):** Findings obtained on small teaching models (GfSE collection) do not transfer to a large multi-layer model (Apollo 11) without a semantic navigation layer.

Approach

Use case	Example task on Apollo 11	Ground truth / metric
Abfragen (query)	“Which technical components satisfy requirement X?”	exact set match vs. model query
Validierung (validation)	“Is package P well-formed and CoSMA-conformant?”	tool validation suite results
Verifizierung (verification)	“Is mass budget requirement R satisfied by the current values?”	expression evaluation in tool
Korrektur (correction)	repair injected faults (dangling ref., wrong type, broken <code>satisfy</code>)	fault-repair rate, no collateral change
Erstellen (creation)	add missing technical component with ports + satisfy links	syntactic validity + expert rubric

Table 1: Use-case catalogue (to be refined).

[TODO 150 words prose around the table]

Evaluation Methodology

[TODO 150 words]

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cache layer of the bridge and for querying inside the tool rather than through the REST API; PDF at [bib/papers/ahlbrecht2025mbsqle.pdf](#).

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